Curriculum Vitae

Ivan Novikov
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Education

- 10/2021 – 12/2025: PhD in Applied Mathematics, Université Paris Dauphine-PSL.
  Thesis: "Zero-Sum Stochastic Games with Vanishing Stage Duration and Public Signals", under the supervision of Guillaume Vigeral.
  Defense Date: December 3, 2025.
  Committee: Sylvain Sorin (chair), Catherine Rainer (referee), Eran Shmaya (referee), Pierre Cardaliaguet (examiner), Bruno Ziliotto (examiner), Guillaume Vigeral (advisor).
- 09/2020 – 09/2021: M.Sc. in Optimization, Université Paris-Saclay, with highest honors.
- 09/2016 – 08/2020: B.Sc. in Mathematics, National Research University Higher School of Economics (Moscow).

Positions

- 10/2026 – 09/2028: Postdoctoral Fellow, School of Data Science, The Chinese University of Hong Kong, Shenzhen. Advisor: Xiao Li.
- 09/2025 – 08/2026: Half-time Temporary Teaching and Research Assistant (Demi-ATER), Université Paris 1 Panthéon-Sorbonne.
- 10/2024 – 08/2025: Half-time Temporary Teaching and Research Assistant (Demi-ATER), CEREMADE, Université Paris Dauphine-PSL.

Publications

1. *Asymptotic Value in Zero-Sum Stochastic Games with Vanishing Stage Duration and Public Signals*. To appear in **International Journal of Game Theory** (2026). `https://arxiv.org/abs/2403.07467`.
2. *Zero-Sum State-Blind Stochastic Games with Vanishing Stage Duration*. **Dynamic Games and Applications** 15, 1094–1115 (2025). `https://doi.org/10.1007/s13235-024-00599-y`.
3. *Percolation of Three Fluids on a Honeycomb Lattice*. **Monatshefte für Mathematik** 199, 611–626 (2022). `https://doi.org/10.1007/s00605-022-01753-7`.
4. *Feynman Checkers: the Probability to Find an Electron Vanishes Nowhere Inside the Light Cone*. **Reviews in Mathematical Physics** 34, 2250020 (2022). `https://doi.org/10.1142/S0129055X22500209`.
5. *Cutting Polygons Composed of Equal Rectangles into Similar Rectangles*. **Mathematical Notes** 110, 726–731 (2021). `https://doi.org/10.1134/S0001434621110092`.

Preprints, submitted, or work in progress

6. *Strategies in POMDPs with Stage Duration*. Submitted (2026). `https://arxiv.org/abs/2603.16055`.

Conference Talks

- *Première rencontre nationale du RT Optimisation* (Villeurbanne, France), 26/11/2025 – 28/11/2025, poster.
- *30 Years of Game Theory at Institut Henri Poincaré* (Paris, France), 06/10/2025 – 10/10/2025, poster.

- *16th Viennese Conference on Optimal Control and Dynamic Games* (Vienna, Austria), 15/07/2025 – 18/07/2025, 25-minute presentation.
- *Workshop "Dynamic Games"* (Fréjus, France), 26/05/2024 – 28/05/2024, 45-minute presentation.
- *Journées SMAI MODE 2024* (Lyon, France), 27/03/2024 – 29/03/2024, 30-minute presentation.
- *Journées annuelles 2023 du GdR MOA* (Perpignan, France). 18/10/2023 – 20/10/2023, 30-minute presentation.
- *The 2023 Alpine Game Theory Symposium* (Grenoble, France). 26/06/2023 – 30/06/2023, lightning talk + poster presentation.

Seminar Talks

- *Junior Game Theory Seminar*, Institut Henri Poincaré (Paris, France), 29/06/2026.
- *Séminaire parisien de Théorie des Jeux*, Institut Henri Poincaré (Paris, France), 30/03/2026.
- *Junior Game Theory Seminar*, Institut Henri Poincaré (Paris, France), 26/05/2025.
- *Matinée du Ceremade*, Université Paris-Dauphine (Paris, France), 14/01/2025.
- *Tropical seminar*, École polytechnique (Palaiseau, France), 05/12/2024.
- *Young Researchers Seminar Days* (Saint-Pierre-Canivet, France), 11/06/2024.
- *Junior Game Theory Seminar*, Institut Henri Poincaré (Paris, France), 11/03/2024.
- *Combinatorial and number-theoretic problems on lattices*, Higher School of Economics (Moscow, Russia, participated online), 16/02/2023.
- *I.Kh. Sabitov's Seminar*, Moscow State University (Moscow, Russia), 25/10/2019.

Scholarships, awards

- 2020–2021: Sophie Germain Master's Scholarship (FMJH).

Teaching

1) Place: Université Paris 1 Panthéon-Sorbonne, B.Sc. in Mathematics and Computer Science Applied to the Social and Human Sciences.

- Spring 2026: (3rd year) *Optimization* (tutorials, 36h, in French).
- Fall 2025: (3rd year) *Analysis S5* (tutorials, 42h, in French).
- Fall 2025: (1st year) *Computer Science S1* (practical work, 24h, in French).

2) Place: Université Paris Dauphine-PSL, B.Sc. in Applied Mathematics.

- Spring 2025: (3rd year) *Differential Calculus and Numerical Optimization* (tutorials and practical work, 41h, in French).
- Fall 2024: (2nd year) *Linear Algebra 3* (tutorials, 51h, in French).
- Fall 2023: (3rd year) *Differential Calculus and Optimization* (tutorials, 44h, in French).

Skills

Computer skills: Python, Wolfram Mathematica, Matlab.
Language skills: English (fluent), French (fluent), Russian (mother tongue).